

Lecture 5: Sector coupling and transmission

Energy Economics of the Green Transition

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Today

1. Where we left things: the two tails of the balancing problem
2. Trade over wires
3. Electrification and heat
4. A workshop on Ruhnau et al. (2020): do heat pumps help integrate wind?

Where we left things

Where we left things

- **By levelised cost**, wind and solar are the cheapest plants we can build (lecture 4).
- **Their value falls as they grow** (lecture 3). The problem is not what they cost but when they produce: every hour, supply must meet demand.
- **Four ways to balance the hour:** flexible backup, storage, flexible demand, trade.
- **Storage** can work on both tails of the problem, at a cost set by the duration: hours (batteries), days (pumped hydro), seasons (reservoir hydro).
- **Today:** trade, and flexible demand from electrified heat.

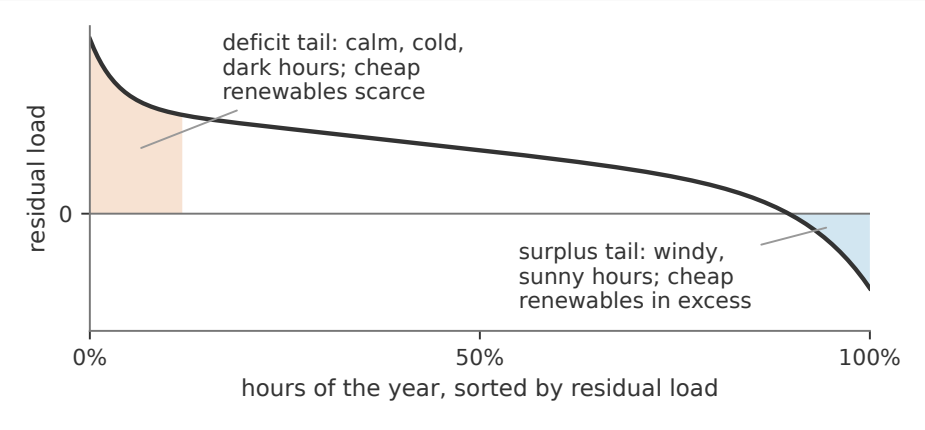
The hourly balance, and what can move

Every hour, in *Energy System Models*' notation:

$$\underbrace{\gamma_h \bar{q}_w}_{\text{wind and solar}} + \underbrace{q_{g,h}}_{\text{dispatchable plants}} + \underbrace{g_h^d - g_h^c}_{\text{storage}} = \underbrace{D_h}_{\text{demand}} + \underbrace{x_h}_{\text{net exports}}$$

- **Trade**, x_h: a surplus is sold to the neighbours, a shortfall bought from them.
- **Flexible demand**, D_h: consumption moves to the hours when power is plentiful. Heat is the case, if it can wait.
- Today asks of both: which part of the problem does it relieve?

The two tails of the balancing problem



Stylised: load minus wind and solar for every hour of the year, sorted from highest to lowest (lecture 4's residual load duration curve).

Trade over wires

Trade moves the surplus in space

The balance of lecture 4, one per zone z:

$$\sum_{g \in \mathcal{G}_z} q_{g,h} + \underbrace{\sum_{\ell:\, b_\ell = z} (1 - \phi_\ell) f_{\ell,h}}_{\text{imports, net of losses}} = D_{z,h} + \underbrace{\sum_{\ell:\, a_\ell = z} f_{\ell,h}}_{\text{exports}} \qquad (\lambda_{z,h})$$

- Every zone has its own balance, and its own price $\lambda_{z,h}$.
- A corridor ℓ carries a flow $f_{\ell,h}$ from zone a_ℓ to zone b_ℓ, and loses the share ϕ_ℓ of it on the way.
- Storage moves a surplus to another hour; trade moves it to another place, where residual demand is high.

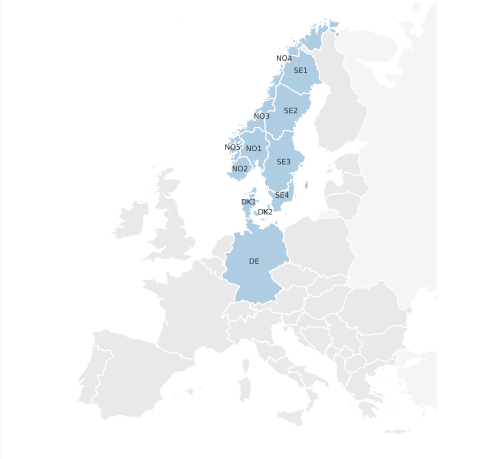
Energy System Models, §6.3. Wind and solar are among the $q_{g,h}$ here.

Question

Can trade relieve the two tails of the balancing problem?

1. only the deficit tail
2. only the surplus tail
3. both
4. neither

Europe's market: bidding zones



Energy System Models, Fig. 6.1: the bidding zones of the day-ahead market; ESM's twelve highlighted.

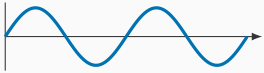
AC versus DC

DC, direct current



- Steady, one way
- Converters at each end set the flow: **controllable**
- Long sea cables; links between grids not in step

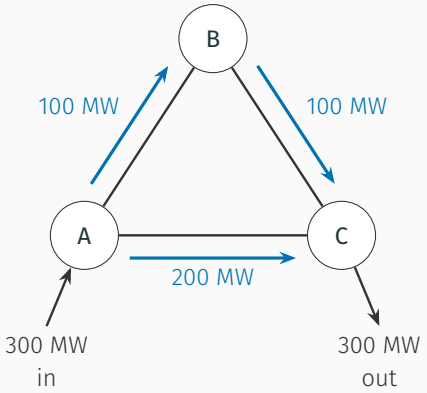
AC, alternating current



- Swings 50 times a second; most of the grid
- Power takes every path, more the shorter ones: **not controllable**
- A line's flow depends on the whole network

The voltage on a wire over time, stylised.

AC and DC



Energy System Models, Fig. 6.3, redrawn: 300 MW from A to C over three identical AC lines.

- **DC:** the operator controls the flow, anywhere between zero and the capacity.
- **AC:** the split follows physics. Power takes every path, in inverse proportion to the paths' reactance, so B carries a *loop flow* of a trade it is no part of.
- **The model** keeps one linear constraint per AC loop.

A line, to an economist: four numbers

A line ℓ carries power from zone a_ℓ to zone b_ℓ; a line used both ways counts twice, once per direction.

- **Capacity** $\bar{f}_\ell$, MW: what the market may trade on it, the *net transfer capacity*, set below the thermal rating for security (DK1–Germany: 2.5 GW).
- **Losses** ϕ_ℓ: resistance turns flow into heat; ESM charges 3% per DC crossing.
- **Cost:** long lines are DC, and under the sea DC is the only option.
- **Control:** can an operator set the flow? On DC, yes; on AC, no.

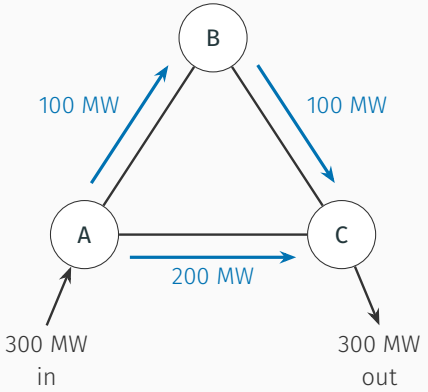
AC overhead line	442	EUR/MW per km
DC submarine cable	1008	EUR/MW per km
DC converter station pair	166	thousand EUR/MW

Energy System Models, §6.2 and Table 3; a 40-year lifetime.

The model: zones and corridors

min	$\sum_h \sum_z \sum_{g \in \mathcal{G}_z} c_g \, q_{g,h}$	cost	
s.t.	$\sum_{g \in \mathcal{G}_z} q_{g,h} + \sum_{\ell:\, b_\ell = z} (1 - \phi_\ell) f_{\ell,h} = D_{z,h} + \sum_{\ell:\, a_\ell = z} f_{\ell,h}$	zonal balance	$(\lambda_{z,h})$
	$0 \le q_{g,h} \le \gamma_{g,h} \, \bar{q}_g$	plant capacity	$(\underline{\mu}_{g,h},\ \overline{\mu}_{g,h})$
	$0 \le f_{\ell,h} \le \bar{f}_\ell$	corridor capacity	$(\underline{\omega}_{\ell,h},\ \overline{\omega}_{\ell,h})$
	$\sum_{\ell \in C} \sigma_{C\ell} \, x_\ell \, f_{\ell,h} = 0$	each AC loop C	$(\zeta_{C,h})$

The loop flow, in the model's equations



Energy System Models, Fig. 6.3, redrawn.

No losses, and every line has reactance x:

$$\begin{aligned}
\text{balance A:} \quad & 300 = f_{AB} + f_{AC} \\
\text{balance B:} \quad & f_{AB} = f_{BC} \\
\text{balance C:} \quad & f_{AC} + f_{BC} = 300 \\
\text{loop A}\to\text{B}\to\text{C}\to\text{A:} \quad & x f_{AB} + x f_{BC} - x f_{AC} = 0
\end{aligned}$$

- **The balances** leave one thing open: how much goes through B.
- **The loop constraint** settles it: $f_{AB} = f_{BC} = 100$ and $f_{AC} = 200$.
- **Three DC cables instead:** no loop constraint, and the operator picks the split.

A DC cable: the flow condition

Differentiate the Lagrangian with respect to the flow on a DC cable from a to b:

$$(1 - \phi_\ell)\,\lambda_{b,h} - \lambda_{a,h} + \underline{\omega}_{\ell,h} - \overline{\omega}_{\ell,h} \;=\; 0$$

- **Nothing sent:** $\underline{\omega}_{\ell,h} = \lambda_{a,h} - (1 - \phi_\ell)\lambda_{b,h} \geq 0$. The far price, net of losses, does not cover the home price.
- **Used, below capacity:** $\lambda_{b,h} - \lambda_{a,h} = \phi_\ell\,\lambda_{b,h}$. The prices differ by the *loss wedge* and nothing else.
- **Congested:** $\overline{\omega}_{\ell,h} = (1 - \phi_\ell)\lambda_{b,h} - \lambda_{a,h} > 0$. The prices decouple; the spread net of losses is what one more MW of cable is worth in that hour.

What a MW of cable is worth

- **One more MW of corridor** lowers the year's cost by the sum of its hourly congestion prices (the envelope theorem):

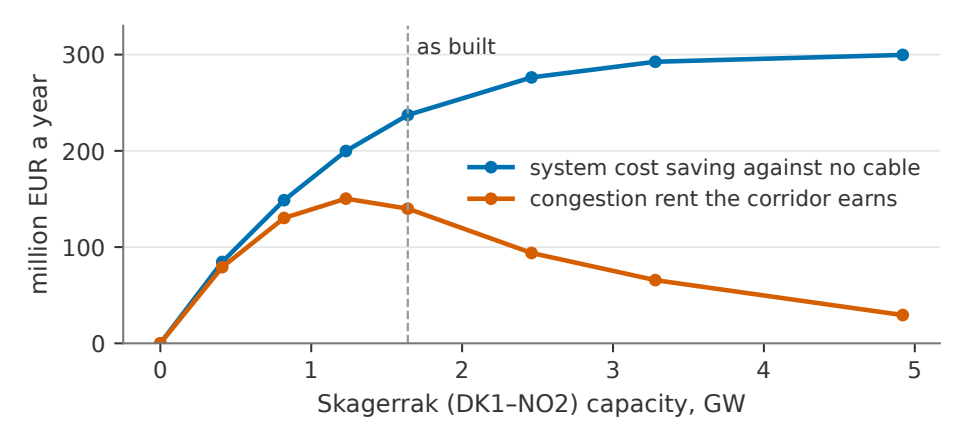
$$\frac{\partial C^*}{\partial \bar{f}_\ell} = -\sum_h \overline{\omega}_{\ell,h}.$$

  Build it while $\sum_h \overline{\omega}_{\ell,h}$ exceeds the annuity of a MW of cable.

- **The rent the owner collects**, flow times spread, splits in two:

$$R_\ell = \sum_h f_{\ell,h}\left(\lambda_{b,h} - \lambda_{a,h}\right) = \underbrace{\sum_h \overline{\omega}_{\ell,h}\,\bar{f}_\ell}_{\text{scarcity rent}} + \underbrace{\sum_h \phi_\ell\,\lambda_{b,h} f_{\ell,h}}_{\text{value of the losses}}.$$

The value of another gigawatt

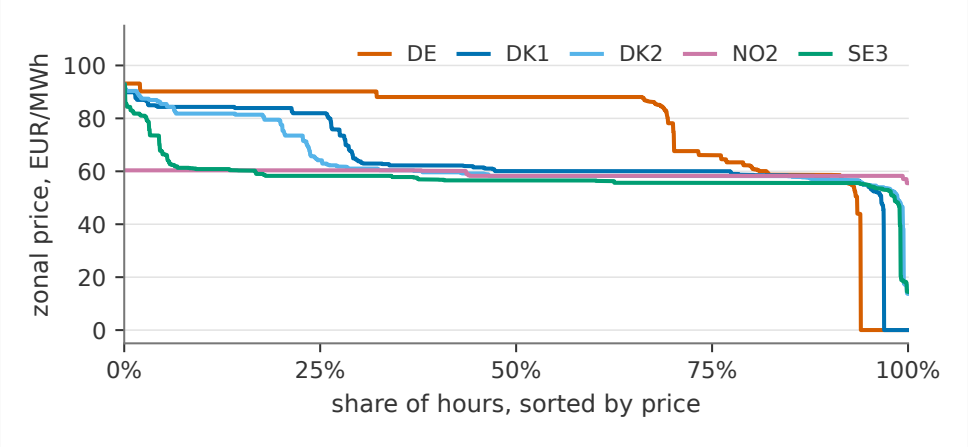


Energy System Models, Fig. 6.8, redrawn: the Skagerrak swept from zero to three times its size, 2024. Gross of the cable's cost.

When does trade pay?

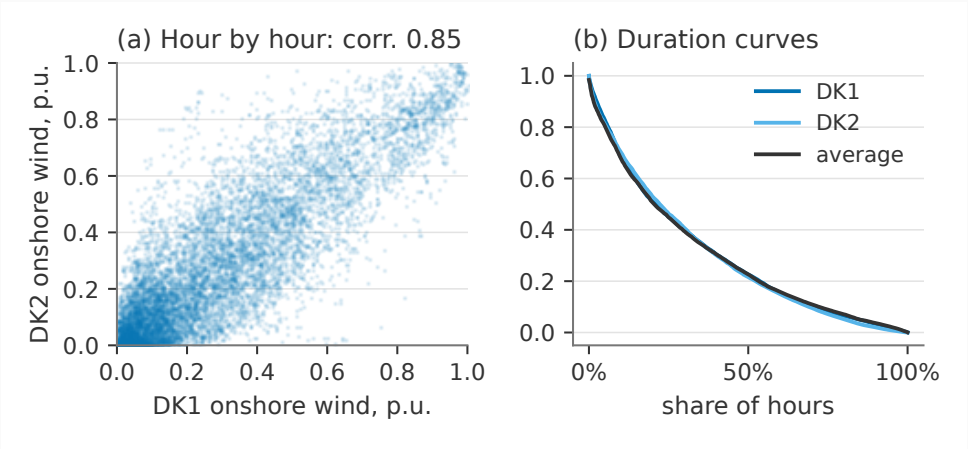
- A wire earns the price spread, so trade pays only where prices differ.
- **In level:** different marginal costs, Nordic water against German gas and coal. Trade moves cheap energy to costly places.
- **In timing:** residual loads that do not move together, because the weather or the demand differs. Trade moves a surplus to where there is a deficit *in the same hour*.
- The balancing problem needs the second: a neighbour whose windless week is not ours.

Denmark between two neighbours



Energy System Models, Fig. 6.4, redrawn: price duration curves of the five zones that face Denmark, the model of 2024.

Neighbours with the same weather



Energy System Models, Fig. 6.7, redrawn: DK1 and DK2 onshore wind, hourly, 2024.

What trade does to the two tails

- **Both tails, when the neighbour differs:** imports cover the deficit tail, exports drain the surplus tail.
- **In space, not in time:** a calm week in Denmark is covered from full Norwegian lakes, which no four-hour battery can do.
- **Not when everyone is short at once:** a still, cold week over all of northern Europe, or a sunny noon everywhere.
- **Only up to the wire's size:** the hours it is full are the hours it earns, and the hours it cannot help more.

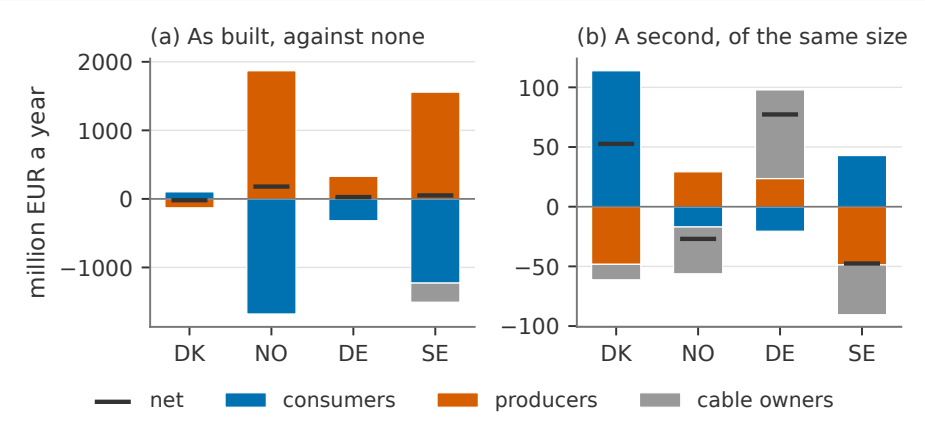
Who wins and who loses from a cable

	low-price, exporting zone	high-price, importing zone
price	rises	falls
consumers	lose	win
producers	win	lose
cable owner	collects the congestion rent	
net	positive: the system cost saving	

Energy System Models, Table 4.

The sum is the gain from trade. Two of the four cells are losses, and they are concentrated.

The Skagerrak: who gains



Energy System Models, Fig. 6.9, redrawn: the change in consumer surplus, producer profit and each country's half of the rents, 2024. The two panels have different scales.

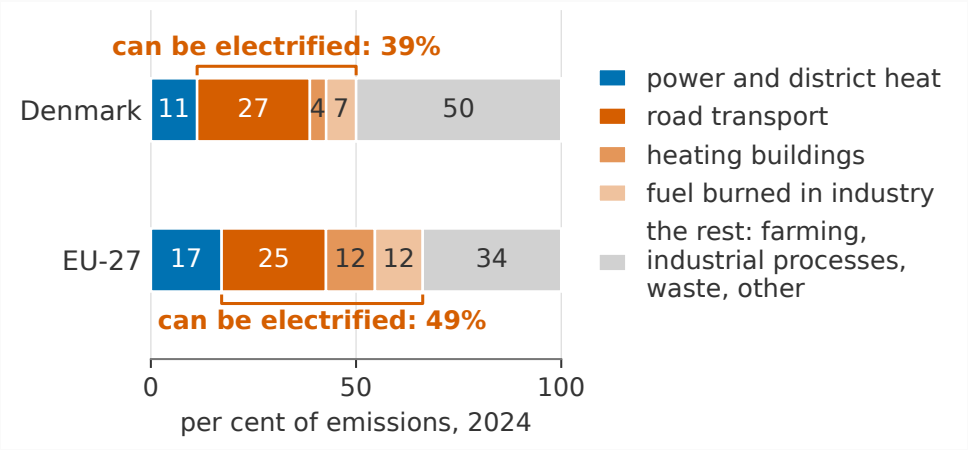
Who pays for a cable?

- **Split it by ownership:** the Nordic convention, half each (Skagerrak: Energinet and Statnett). Simple, and unrelated to who gains.
- **Let the beneficiaries pay:** FERC's Order 1000 in the US; in the EU, cross-border cost allocation for projects of common interest, decided by the regulators or by ACER.
- **Let the cable pay for itself:** a merchant line keeps its rent, and stops at the hump.
- **Bargain over the surplus:** cooperative game theory, such as the Shapley value.

Energy System Models, §6.7. Hogan (2018) is the primer; Joskow and Tirole (2005) on merchant lines.

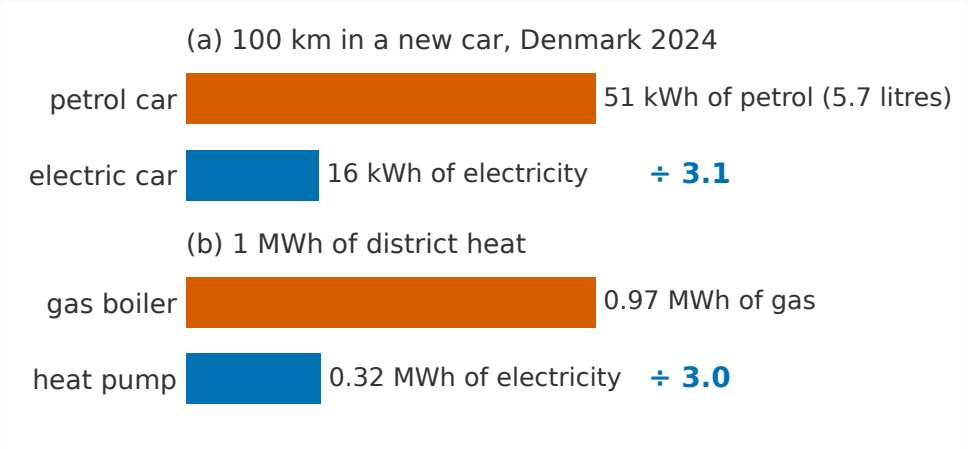
Electrification and heat

Why electrify? The emissions are outside the power sector



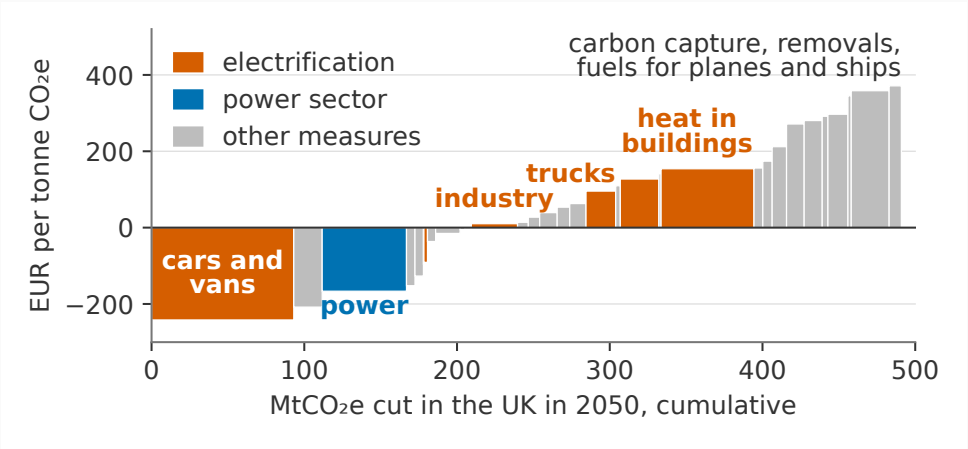
Eurostat, greenhouse-gas inventory (env_air_gge), 2024; land use and international aviation and shipping excluded.

Why electrify? A third of the energy



(a) EEA, CO_2 monitoring of new cars (CC BY 4.0), test-cycle averages; petrol at 32 MJ/l. (b) *Generation Technologies*, 2025.

Why electrify? The cheap end of the cost curve



Climate Change Committee (2025), *The Seventh Carbon Budget*, Fig. 4.6, redrawn; Open Government Licence v3.0; 1.15 EUR/GBP.

Why electrify? A heat pump moves heat

Per MWh of district heat, 2025 costs:

	gas boiler	heat pump
energy in	about 1 MWh of gas	about 0.3 MWh of electricity (COP 3.15)
cost, EUR	37 + carbon (53 at 80 EUR/t)	50 (electricity at 60 EUR/MWh)
CO_2, t	0.19	0 with clean power; 0.11 with gas; 0.30 with coal

- Switching costs 68 EUR per tonne avoided if the electricity is clean, 161 if it comes from gas.
- With coal at the margin, the heat pump emits *more* than the boiler.

Generation Technologies: district-scale heat pump and gas boiler, both at the heat pump's 35% utilisation.

Flexible demand: a margin, or a new load?

- **A margin:** heat is cheap to store, in a tank, in the pipes, in a building's walls; and district heating can switch to a boiler when power is costly. So electrified heat could move to the hours when power is plentiful.
- **A new load:** heat demand is driven by the weather. It peaks in the cold hours, when a heat pump's COP is lowest and the wind may be calm.
- Which it is depends on what stands behind the heat pump: a store, a second fuel, or nothing.

Question

Can flexible demand relieve the two tails of the balancing problem?

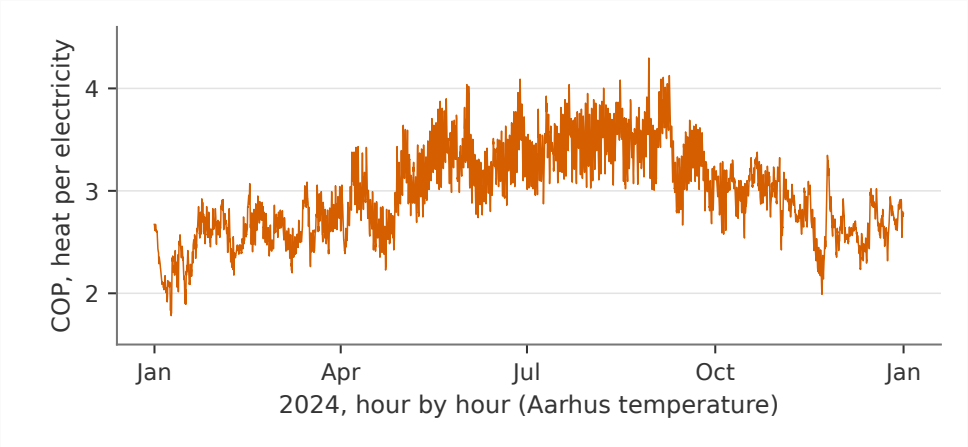
1. only the deficit tail
2. only the surplus tail
3. both
4. neither

Heat markets are not power markets

	electricity	district heat
reach	a continent, by wire	a town, by pipe
trade	across borders, hour by hour	none beyond the network
storage	costly (lecture 4)	cheap: tanks, pipes, buildings
balance	every second	with hours of slack
price	hourly, from a market	a tariff, from the heat company's costs

The heat pump joins the two: a consumer in one market, a producer in the other.

The heat pump couples the two markets



Energy System Models, Fig. 5.1(b), redrawn. When the pump sets the heat price, $\lambda_h^H = \lambda_h^E / COP_h$.

Workshop: Ruhnau et al. (2020)

Today builds on

***Energy System Models*, section 5:** the heat market and the heat pump, and the rollout (§5.3); and §3.4 for the capture price and the value factor.

Ruhnau et al. (2020), *Heating with wind: Economics of heat pumps and variable renewables*. Sections 1–5.

Optional, on trade: Gonzales et al. (2023).

The question

The hope (the paper's p. 2): heat demand and wind are both higher in winter, and heat is cheap to store, so heat pumps should use wind "over-proportionately" and help integrate it.

heat pumps from 0 to 15% of consumption: their load cost	+21 EUR/MWh
30% wind, 15% heat pumps, wind capacity held: wind's value	+2 to 3 EUR/MWh
30% wind, 15% heat pumps, wind share held: wind's value	about 0

Why so much for the heat pumps, and so little for wind?

Part 1

Setup: two prices, weighted two ways

$$\text{wind's market value (4)} = \frac{\sum_h w_h \lambda_h}{\sum_h w_h} \qquad \text{heat pump load cost (5)} = \frac{\sum_h l_h \lambda_h}{\sum_h l_h}$$

- λ_h the price, w_h wind's output, l_h the heat pumps' consumption, in hour h.
- **The model:** the planner builds everything except the wind turbines and the heat pumps, whose shares the authors set, and it can shed load.
- **In §3.1–3.2** the heat pumps follow the heat load, whatever the price.
- Keep the paper open at its figures: part 1 reads Figs. 1–7.

What heat pumps cost, and what they do for wind

1. The research question: why it matters; the two opposing forces on wind's value.
2. Heat pumps alone: Fig. 1; new capacity (Fig. 2); load shedding (Figs. 2–3).
3. Heat pumps and wind: Fig. 5's two patterns; the two effects in Figs. 6–7.

Stop point

Why the load cost rises steeply as heat pumps grow; heat pumps raise wind's value at a fixed wind capacity, not at a fixed wind share.

The full questions, the hints and the stretch are on the worksheet.

Solutions 1

Solutions 1, Q1–Q2: the questions

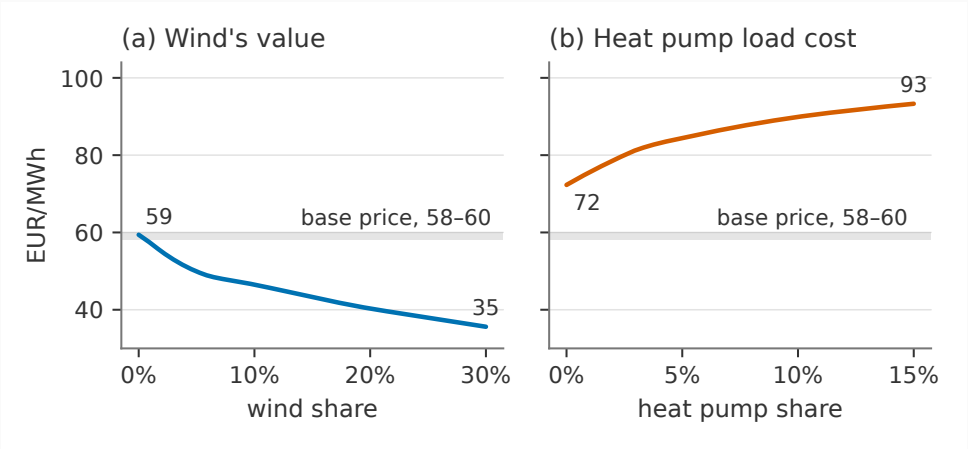
Q1. Why the research questions matter; the two opposing forces that make the effect on wind's value "ambiguous ex ante".

Q2. Fig. 1; what capacity the system adds for heat pumps (Fig. 2); load shedding, and why its cost barely matters (Fig. 3).

Solutions 1, Q1: why it matters, and two forces

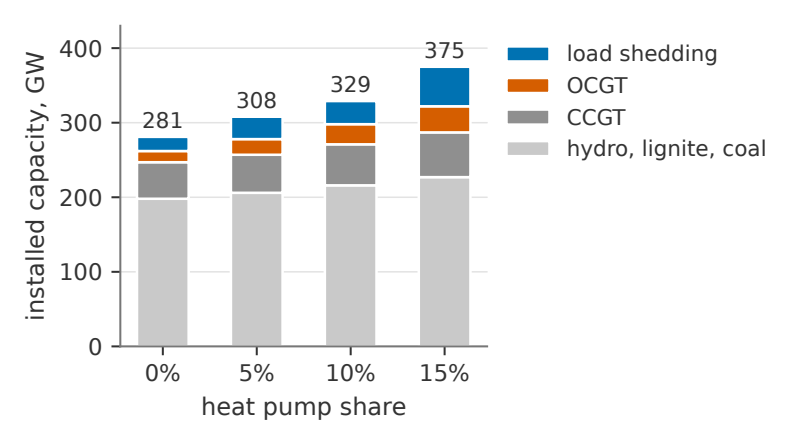
- **Q1. Why it matters:** heat pumps decarbonise heat, wind decarbonises power, and they meet in the power market. Do they help or hinder each other there?
- **Q1. Force one:** heat pumps add demand, more of it in the windy season, which lifts the prices wind sells at.
- **Q1. Force two:** the wind share is held, heat pumps included, so more demand means more wind, which lowers its value again.
- **Q1. Hence** a numerical model: theory cannot say which force wins. Fig. 7 separates them.

Solutions 1, Q2: the mirror image



After Ruhnau et al. (2020), Fig. 1, redrawn schematically on one price axis: (a) without heat pumps, (b) without wind.

Solutions 1, Q2: capacity for the heat pumps' peak



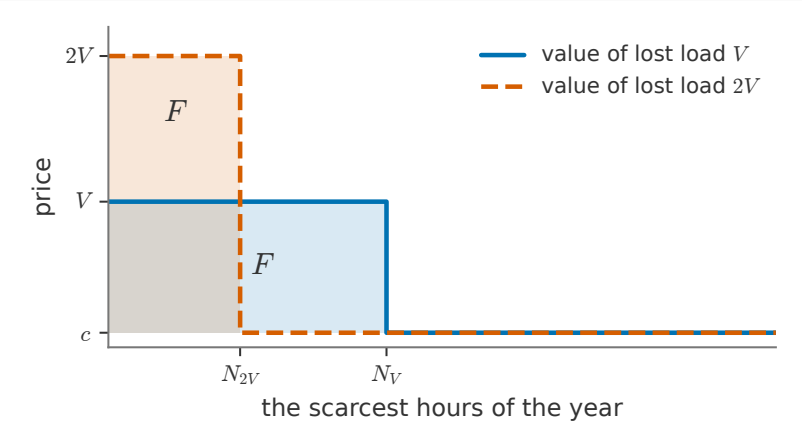
After Ruhnau et al. (2020), Fig. 2, redrawn schematically; hydro, lignite and coal as one block.

Question

The value of lost load doubles, from 1,000 to 2,000 EUR/MWh. What happens to the heat pumps' load cost?

1. its scarcity part roughly doubles
2. it barely moves
3. it falls

Solutions 1, Q2: scarcity pays for the peakers

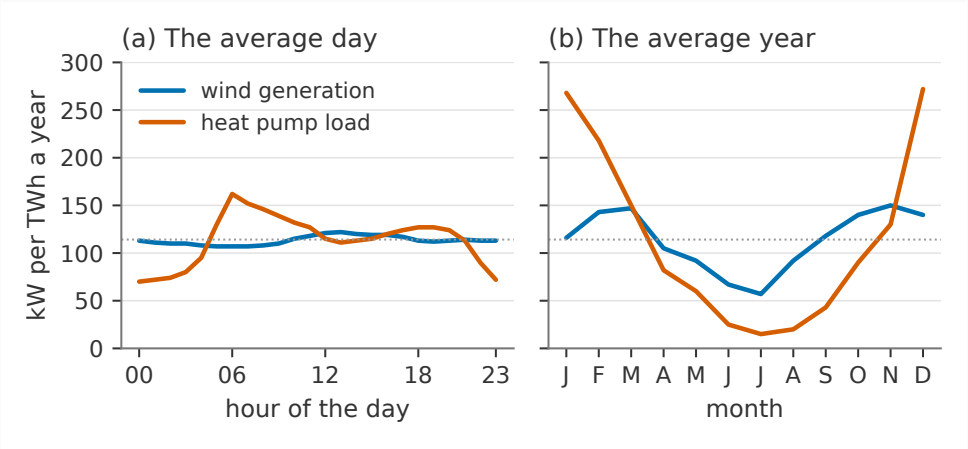


Ours, not the paper's: the scarcest hours of a long-run equilibrium. The last peaking plant is paid by an area equal to its fixed cost *F*.

Solutions 1, Q3: the questions

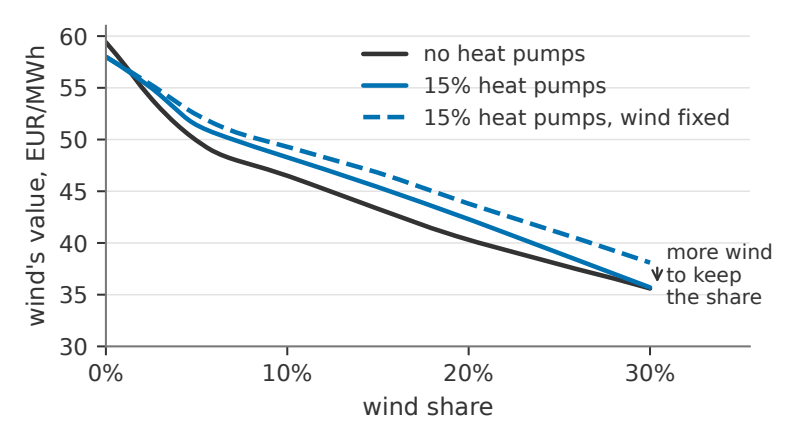
Q3. Fig. 5: the daily and seasonal patterns, and a correlation of 0.11. Figs. 6–7: what heat pumps do to wind's value; what the dashed line holds fixed, and the two effects it separates.

Solutions 1, Q3: winter, but not the same hours



After Ruhnau et al. (2020), Fig. 5, redrawn schematically: kW per TWh of annual energy; dotted, a flat profile.

Solutions 1, Q3: two effects on wind's value



After Ruhnau et al. (2020), Fig. 7, redrawn schematically. Dashed: the wind capacity of no heat pumps.

Part 2

Setup: what takes the cost back

- **The question of §3.3–5:** can the load cost increase of part 1 be brought down, and what does that do for wind?
- **Two remedies** besides wind: “system-friendly” heat pumps (§3.3, Table 1) and thermal storage (§3.4, Eqs. (9)–(10)).
- **Other flexibility** the planner builds: interconnectors and pumped hydro.
- Part 2 reads Figs. 8–13.

Part 2 is self-contained: what it needs from part 1 is at its top on the worksheet.

What takes the cost back, and what it does for wind

1. System-friendly heat pumps: what the term means; the load cost (Fig. 8), and why little for wind.
2. Flexible heat pumps: what flexibilization means; the load cost (Fig. 9), and why little for wind (Figs. 10–12).
3. What it adds up to: Fig. 13; the answer to the title question.

Stop point

Both remedies lower the load cost and, with wind, more than offset its increase; neither does much for wind's value.

The full questions, the hints and the stretch are on the worksheet.

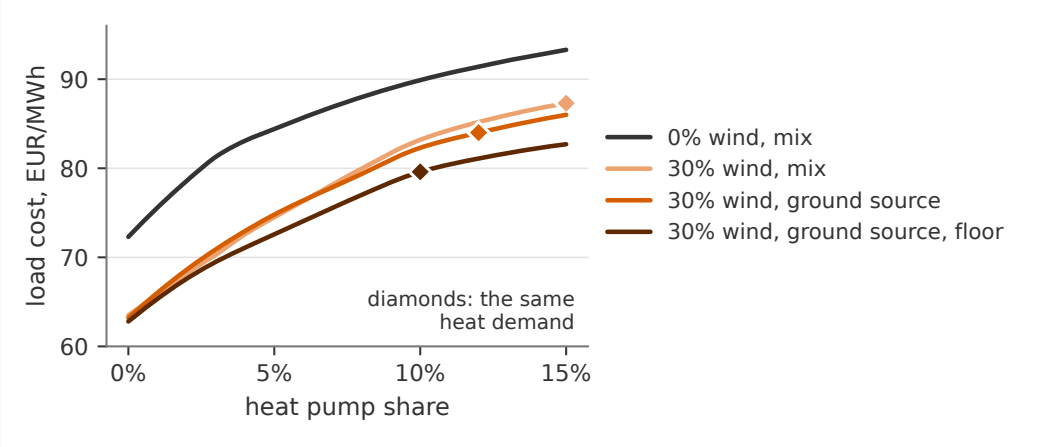
Solutions 2

Solutions 2, Q1–Q2: the questions

Q1. What “system-friendly” means; what it does to the load cost (Fig. 8), and why it does little for wind.

Q2. What “flexibilization” means; what it does to the load cost (Fig. 9), and why it does little for wind (Figs. 10–12).

Solutions 2, Q1: a smaller, steadier load

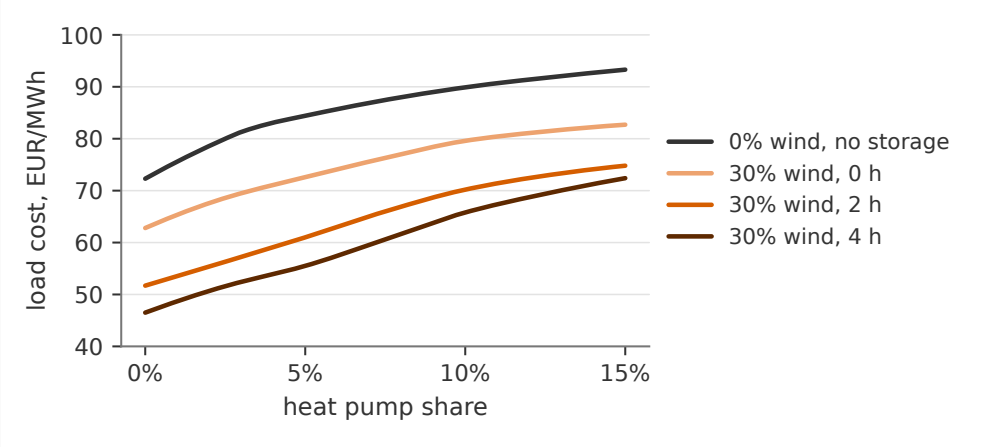


After Ruhnau et al. (2020), Fig. 8, redrawn schematically; the diamonds mark the same heat demand.

Solutions 2, Q1: why so little for wind

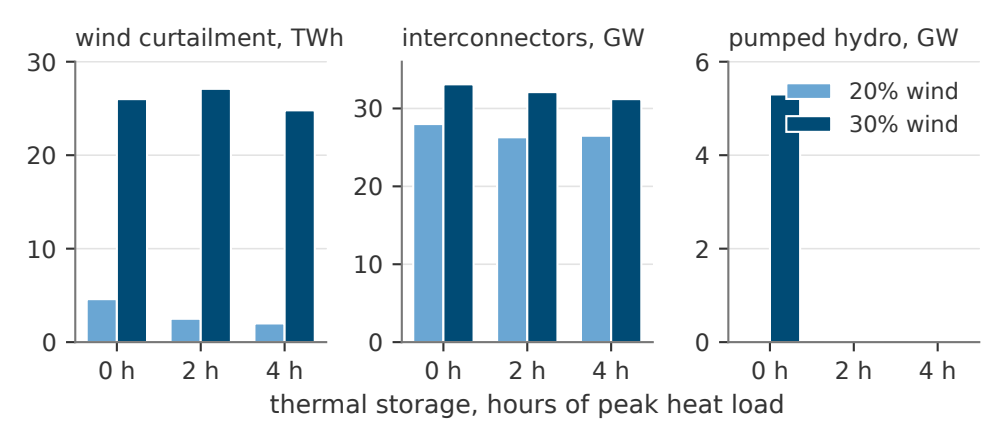
- **Q1. What a steadier load avoids** is its peaks: cold hours with scarcity prices.
- **Q1. Those hours are mostly calm:** wind produces little in them.
- **Q1. In the windy hours** where wind earns its value, the heat pump load hardly changes, so neither does wind's value ("no significant implications", p. 8).

Solutions 2, Q2: storage that holds hours



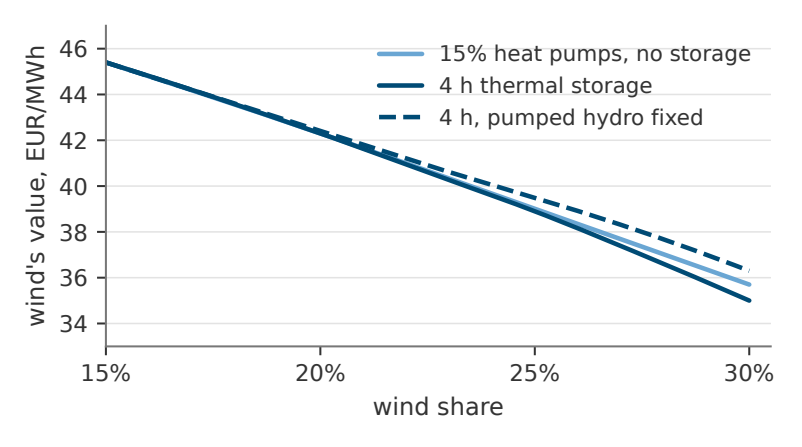
After Ruhnau et al. (2020), Fig. 9, redrawn schematically: ground source with floor heating at 30% wind.

Solutions 2, Q2: storage crowds out other flexibility



After Ruhnau et al. (2020), Fig. 11, redrawn schematically in three panels, at 15% heat pumps.

Solutions 2, Q2: with pumped hydro held, storage helps wind

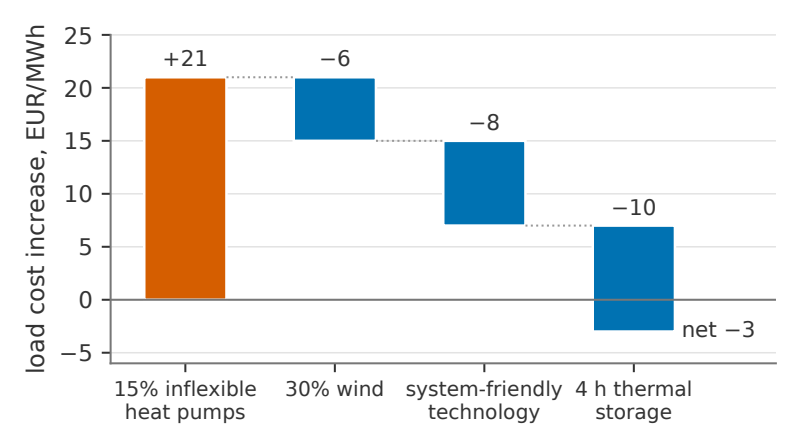


After Ruhnau et al. (2020), Figs. 10 and 12, redrawn schematically over the wind shares where their lines part, at 15% heat pumps.

Solutions 2, Q3: the questions

Q3. Fig. 13 and “overcompensate”. Do heat pumps help integrate wind, or does wind help integrate heat pumps?

Solutions 2, Q3: what it adds up to



After Ruhnau et al. (2020), Fig. 13, redrawn schematically: at a 15% heat pump share.

Solutions 2, Q3: heating with wind

- **Q3. Heat pumps hardly help wind:** they raise its value only at a fixed wind capacity, and neither better technology nor storage changes that.
- **Q3. Wind helps heat pumps:** it lowers their load cost, and with the two remedies more than offsets its increase.
- **Q3. Only if someone chooses the remedies:** on fixed tariffs and standard load profiles nobody is paid to (p. 12); hence settlement on wholesale prices through smart meters.

Wrap-up

What carries forward

1. **Trade relieves both tails when the weather differs across the wire.** A cable is paid by the hours it is full; its value is concave, and its gains are unevenly shared.
2. **A new load has a load cost,** the mirror of a capture price. Heat pumps that follow the heat load consume in the scarcest hours, and cost more the more there are; wind, steadier heat pumps and storage bring it back down.
3. **Flexibility competes with flexibility.** Heat pumps do little for wind once wind grows with them, and thermal storage crowds out cables and pumped hydro.

Lecture 6 lets the planner build the whole fleet: investment.

Before lecture 6 (Monday 12 October): *Energy System Models*, section 7; Brown et al. (2025). Full solutions to today's worksheet go up after class.

References i

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